# MULTIPARTITION SPACES, KIRWAN'S EQUIVARIANT PERFECTION, AND HAUSEL'S FORMULA

Abstract. These notes give a self-contained account of the geometric part of Section 6 of [Hik]. For a quiver with a loop at every vertex and a multipartition $m$, we define the multipartition quiver $\Gamma_m$, the multipartition space $E_m$ and its open subset $U_m$, the torus $T_m$, the toric hyperKähler variety $Y_m$, and the action of the group $W_m$. We then prove [Hik, Proposition 6.2] as an instance of Kirwan's equivariant perfection argument: the decomposition of $E_m$ by the set of unreachable slots is exactly Hesselink's stratification for the determinant character, in the terminology of Hoskins [Hos]. Finally we recall Hausel's formula and explain, following [Hik, Corollary 6.3], how Proposition 6.2 computes the ratio of its two sums.

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## Contents

Conventions 1
1. Multipartition spaces 2
2. Proposition 6.2 via Kirwan's equivariant perfection 8
3. Hausel's formula and Corollary 6.3 13
References 17

## Conventions

Equation, proposition and corollary numbers of the form (6.1), Proposition 6.2, Corollary 6.3 refer to [Hik]. All other numbers are cross-references within this note and are hyperlinks to the corresponding statement or equation.

Cohomology has complex coefficients, and equivariant cohomology is Borel equivariant cohomology. All graded vector spaces below are concentrated in even degrees, finite-dimensional in each degree and bounded below; for such a space we put

$$\mathrm{Hilb}(M;q) = \sum_d \dim M^{2d}\, q^d.$$

Thus $q$ records *half* the cohomological degree. If $X$ has no odd cohomology, its Poincaré polynomial is $P_X(q) = \mathrm{Hilb}(H^*(X);q)$, and $P_\emptyset = 0$.

For geometric invariant theory of a linear action on an affine space with respect to a character we follow the terminology and notation of Hoskins [Hos, §2]. In §1.7 and Section 2, $\lambda$ denotes a one-parameter subgroup; elsewhere $\lambda = (\lambda^i)$ denotes a tuple of partitions.

In [Hik] the polynomials $h_m$ defined below bound the graded pieces of a filtration of the Coulomb-side ring $B_0$ [Hik, Proposition 5.2]; Hausel's formula shows that the bound is sharp [Hik, Proposition 6.4, Corollary 6.6]. We do not revisit that part of the argument.

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## 1. Multipartition spaces

1.1. **The quiver.** Let $Q = (Q_0, Q_1)$ be a finite quiver with at least one loop at every vertex. For an arrow $a \in Q_1$ we write $t(a)$ and $h(a)$ for its tail and head. Fix one loop $\ell_i$ at each vertex $i$, and let $Q'$ be the quiver obtained from $Q$ by deleting the chosen loops. Put

$$n_{ij} = \#\{a \in Q_1' : t(a) = i,\ h(a) = j\}, \qquad c_{ij} = n_{ij} + n_{ji}. \tag{1.1}$$

The matrix $(c_{ij})$ is symmetric with nonnegative integer entries, and $c_{ii} = 2n_{ii}$ is even. Fix a framing vector $w \in \mathbb{Z}_{\geq 0}^{Q_0}$. For $v \in \mathbb{Z}_{\geq 0}^{Q_0}$ put

$$C(v) = \tfrac{1}{2}\sum_{i,j} c_{ij} v_i v_j = \sum_{a \in Q_1'} v_{t(a)} v_{h(a)}. \tag{1.2}$$

1.2. **Multipartitions.**

**Definition 1.1.** A *multipartition* is a family $m = (m_{i,s})_{i \in Q_0,\, s \geq 1}$ of nonnegative integers, only finitely many of which are nonzero. Its *dimension vector* $\|m\| \in \mathbb{Z}_{\geq 0}^{Q_0}$ is $\|m\|_i = \sum_s s\, m_{i,s}$, and $m$ is called a multipartition of $v$ if $\|m\| = v$. Let $\mathcal{M}$ be the set of all multipartitions (of all dimension vectors) and $\mathcal{M}(v) \subset \mathcal{M}$ the finite subset of multipartitions of $v$. The empty multipartition $\emptyset$ is the unique multipartition of 0.

Equivalently, $m$ is a $Q_0$-tuple $\lambda = (\lambda^i)_{i \in Q_0}$ of partitions, where $m_{i,s}$ is the number of parts of $\lambda^i$ equal to $s$; we write $m = m(\lambda)$ and use both descriptions.

**Definition 1.2.** For $k, m \in \mathcal{M}$ put

$$l_i(m) = \textstyle\sum_s m_{i,s} \text{ (the number of parts at } i), \qquad l(m) = (l_i(m))_i, \qquad |l(m)| = \textstyle\sum_i l_i(m), \tag{1.3}$$

$$w(m) = w \cdot l(m) = \textstyle\sum_i w_i\, l_i(m), \qquad (q)_m = \prod_{i,s} \prod_{j=1}^{m_{i,s}} (1 - q^j), \tag{1.4}$$

$$b(k,m) = \sum_{i,j \in Q_0} \sum_{s,t \geq 1} c_{ij} \min(s,t)\, k_{i,s}\, m_{j,t}, \qquad D_m = \tfrac{1}{2}\, b(m,m). \tag{1.5}$$

The form $b$ is bilinear and symmetric. For partitions put $\langle \mu, \nu \rangle = \sum_{a,b} \min(\mu_a, \nu_b)$; then

$$D_{m(\lambda)} = \tfrac{1}{2}\sum_{i,j} c_{ij} \langle \lambda^i, \lambda^j \rangle = \sum_{a \in Q_1'} \langle \lambda^{t(a)}, \lambda^{h(a)} \rangle \in \mathbb{Z}_{\geq 0}. \tag{1.6}$$

Multipartitions are added entrywise, and $\|k + m\| = \|k\| + \|m\|$. We write $k \leq m$ if $k_{i,s} \leq m_{i,s}$ for all $(i,s)$; then $m - k \in \mathcal{M}$. Bilinearity gives

$$D_{k+m'} = D_k + D_{m'} + b(k, m'), \qquad w(k + m') = w(k) + w(m'). \tag{1.7}$$

1.3. **The multipartition quiver and slots.**

**Definition 1.3.** Let $m \in \mathcal{M}$ and put $\Omega_m = \{(i,s,\alpha) : i \in Q_0,\ s \geq 1,\ 1 \leq \alpha \leq m_{i,s}\}$. The *multipartition quiver* $\Gamma_m$ has vertex set $\Omega_m \sqcup \{\infty\}$. The vertex $\infty$ is the *framing vertex*; the other vertices, i.e. the elements of $\Omega_m$, are called *slots*. A slot $d = (i,s,\alpha)$ has *color* $i(d) = i$, *size* $s(d) = s$ and *type* $(i,s)$. For distinct slots $d \neq e$ put

$$b_{de} = c_{i(d)i(e)} \min(s(d), s(e)) = b_{ed}. \tag{1.8}$$

The arrows of $\Gamma_m$ are:

- for every ordered pair $(d,e)$ of distinct slots, $b_{de}$ arrows $d \to e$, labeled $(d \to e, p)$ with $1 \leq p \leq b_{de}$;
- for every slot $d$, $w_{i(d)}$ *framing arrows* $\infty \to d$, labeled $(\infty \to d, p)$ with $1 \leq p \leq w_{i(d)}$.

There are $m_{i,s}$ slots of type $(i,s)$ and $|l(m)|$ slots in all; the sizes of the slots of color $i$ are the parts of $\lambda^i$. By (1.8), the full subquiver on the slots is *symmetric*: it has as many arrows $d \to e$ as $e \to d$. There are no arrows into $\infty$ and no loops.

The group

$$W_m = \prod_{i,s} \mathfrak{S}_{m_{i,s}} \tag{1.9}$$

acts on $\Omega_m$ by $\sigma(i,s,\alpha) = (i,s,\sigma_{i,s}(\alpha))$. Thus $W_m$ is the group of permutations of the slots preserving types, and $b_{\sigma d,\sigma e} = b_{de}$.

For $S \subseteq \Omega_m$ its *type* is the multipartition $\kappa(S) = k \le m$ with $k_{i,s} = \#\{d \in S \text{ of type } (i,s)\}$; the complement $\Omega_m \setminus S$ has type $m-k$. Two subsets lie in the same $W_m$-orbit if and only if they have the same type, and the stabilizer of $S$ is $W_k \times W_{m-k}$. We identify

$$\Omega_m \setminus S \;\cong\; \Omega_{m-k} \tag{1.10}$$

by the type-preserving, order-preserving relabeling of the index $\alpha$; this identification preserves the numbers $b_{de}$. By definition of $b$ in (1.5),

$$b(\kappa(S),\, m-\kappa(S)) = \sum_{d\in S} \sum_{e\in\Omega_m\setminus S} b_{de}, \qquad D_m = \sum_{\{d,e\},\, d\neq e} b_{de} + \sum_{d\in\Omega_m} n_{i(d)i(d)}\, s(d). \tag{1.11}$$

In the second formula the sum is over unordered pairs; the last term comes from the pairs $d = e$ in $\frac{1}{2}b(m,m)$ and vanishes when $Q'$ has no loops.

### 1.4. **The multipartition space.**

**Definition 1.4.** The *multipartition space* $E_m$ is the space of representations of the multipartition quiver $\Gamma_m$ (Definition 1.3) with a one-dimensional space $\mathbb{C}$ at every vertex:

$$E_m = E_m^{\mathrm{int}} \oplus F_m, \qquad E_m^{\mathrm{int}} = \bigoplus_{d\neq e} \mathbb{C}^{b_{de}}, \qquad F_m = \bigoplus_{d\in\Omega_m} \mathbb{C}^{w_{i(d)}}.$$

We write a point as $B = (B', a)$, where $B' = (B^{(p)}_{d\to e}) \in E_m^{\mathrm{int}}$ consists of the components $B^{(p)}_{d\to e}$, $1 \le p \le b_{de}$, for the arrows between slots, and $a = (a_d^{(p)}) \in F_m$ consists of the components $a_d^{(p)}$, $1 \le p \le w_{i(d)}$, for the framing arrows.

Thus $\dim E_m = 2\sum_{\{d,e\}} b_{de} + w(m)$, and $E_\emptyset$ is a point.

### 1.5. **The torus and the cut ideal.**

The torus $T_m = (\mathbb{C}^\times)^{\Omega_m}$ acts on $E_m$ by change of basis at the slots, trivially at $\infty$:

$$t\cdot(B',a): \quad B^{(p)}_{d\to e} \mapsto t_e\, t_d^{-1}\, B^{(p)}_{d\to e}, \qquad a_d^{(p)} \mapsto t_d\, a_d^{(p)}. \tag{1.12}$$

Let $x_d \in H^2_{T_m}(\mathrm{pt})$ be the first Chern class of the character $t \mapsto t_d$, so that

$$P_m := H^*_{T_m}(\mathrm{pt}) = \mathbb{C}[x_d : d \in \Omega_m], \qquad \deg x_d = 2.$$

By (1.12) the weights of $E_m$ are

$$x_e - x_d \;\text{ on each coordinate } B^{(p)}_{d\to e}, \qquad x_d \;\text{ on each coordinate } a_d^{(p)}. \tag{1.13}$$

Since $E_m$ is contractible, $H^*_{T_m}(E_m) = P_m$.

**Definition 1.5.** For a nonempty subset $S \subseteq \Omega_m$, the *cut polynomial* [Hik, (5.3)] is

$$q_S = \prod_{d\in S} x_d^{\,w_{i(d)}} \prod_{d\in S,\; e\notin S} (x_d - x_e)^{b_{de}} \in P_m.$$

The *cut ideal* $J_m \subseteq P_m$ is the ideal generated by all $q_S$, and

$$h_m(q) = \mathrm{Hilb}\left((P_m/J_m)^{W_m}; q\right),$$

where $W_m$ acts on $P_m/J_m$ as in §1.6.

We keep the notation of [Hik]; $q_S$ has nothing to do with the variable $q$. By [Hik, Lemma 6.1], $P_m/J_m$ is finite-dimensional, so $h_m$ is a polynomial. This also follows from Remark 1.12(iv): there $P_m/J_m$ is identified with the cohomology of the toric hyperKähler variety $Y_m$, whose Betti numbers $b_{2k}$ are the $h$-numbers $h_k$, $0 \le k \le g_m$, of a matroid of rank $g_m$ [HS, Corollaries 4.6 and 6.6], where $g_m$ is as in Remark 1.12(ii). Hence $P_m/J_m$ vanishes in degrees above $2g_m$, and $h_m$ has degree at most $g_m$. For $m = \emptyset$ we have $P_\emptyset = \mathbb{C}$, $J_\emptyset = 0$ and $h_\emptyset = 1$.

1.6. **The action of $W_m$.** The arrow labels in Definition 1.3 depend only on the types of the endpoints, so the group $W_m$ of (1.9) acts on $\Gamma_m$, and hence on $E_m$ and on $T_m$:

$$(\sigma B)^{(p)}_{\sigma d \to \sigma e} = B^{(p)}_{d\to e}, \qquad (\sigma a)^{(p)}_{\sigma d} = a^{(p)}_d, \qquad (\sigma t)_{\sigma d} = t_d.$$

These satisfy $\sigma(t \cdot B) = (\sigma t)\cdot(\sigma B)$, so the semidirect product $T_m \rtimes W_m$ acts on $E_m$. (It is the normalizer of the maximal torus in $\prod_{i,s} GL_{m_{i,s}}$.) If $\mathcal{X} \subseteq E_m$ is stable under $T_m \rtimes W_m$, each $\sigma$ gives an equivariant automorphism of $(\mathcal{X}, T_m)$; letting $\sigma$ act by $(\sigma^{-1})^*$ defines a left action of $W_m$ on $H^*_{T_m}(\mathcal{X})$ by graded algebra automorphisms. On $H^*_{T_m}(\mathrm{pt}) = P_m$ it is $\sigma \cdot x_d = x_{\sigma d}$. We record three facts.

(a) Since $W_m$ is finite and coefficients are complex, taking $W_m$-invariants is exact, and

$$H^*_{T_m \rtimes W_m}(\mathcal{X}) = H^*_{T_m}(\mathcal{X})^{W_m}.$$

(b) $\sigma \cdot q_S = q_{\sigma S}$, hence $J_m$ is $W_m$-stable and $W_m$ acts on $P_m/J_m$.

(c) $P_m^{W_m}$ is the tensor product over $(i,s)$ of rings of symmetric polynomials in $m_{i,s}$ variables, so $\mathrm{Hilb}(P_m^{W_m}; q) = 1/(q)_m$, with $(q)_m$ as in (1.4).

1.7. **Stability for the torus action.** We use the notation of [Hos, §2], specialized to the torus $T_m$ acting on $E_m$. Characters and one-parameter subgroups (1-PSs) of $T_m$ are identified with vectors in $\mathbb{Z}^{\Omega_m}$: a character $\chi$ is $t \mapsto \prod_d t_d^{\chi_d}$, a 1-PS $\lambda$ is $\tau \mapsto (\tau^{\lambda_d})_d$, and the pairing is $(\chi, \lambda) = \sum_d \chi_d \lambda_d$. We write characters additively, so $x_d$ is the character $t \mapsto t_d$, and we use the standard norm $\|\lambda\|^2 = \sum_d \lambda_d^2$, which is invariant under $W_m$ and satisfies [Hos, Assumption 2.9]. For $A \subseteq \Omega_m$ let $1_A \in \mathbb{Z}^{\Omega_m}$ have entries 1 on $A$ and 0 elsewhere. Throughout, the character is

$$\rho = \det = 1_{\Omega_m} : \ T_m \to \mathbb{C}^\times, \qquad t \mapsto \prod_d t_d. \tag{1.14}$$

In King's notation for the quiver $\Gamma_m$, this is the stability parameter which is 1 at every slot [Hos, §5.1].

**Definition 1.6** ([Hos, Definition 2.1])**.** A point $B \in E_m$ is *$\rho$-semistable* if there are $n > 0$ and $f \in \mathbb{C}[E_m]$ with $f(t \cdot B'') = \rho(t)^n f(B'')$ for all $t \in T_m$, $B'' \in E_m$, and $f(B) \neq 0$. It is *$\rho$-stable* if moreover its stabilizer is finite and $T_m$ acts on $\{f \neq 0\}$ with closed orbits, for some such $f$. Otherwise $B$ is *$\rho$-unstable*. We write $E_m^{\rho\text{-}s} \subseteq E_m^{\rho\text{-}ss}$ for the open subsets of $\rho$-stable and $\rho$-semistable points, and $E_m /\!/_\rho T_m$ for the GIT quotient.

For $B \in E_m$ let $\mathrm{wt}(B)$ be the set of weights $\chi$ in (1.13) such that the component of $B$ in the $\chi$-weight space is nonzero. Following [Hos, (4)], the *cone of allowable 1-PSs* of $B$ is

$$C_B = \bigcap_{\chi \in \mathrm{wt}(B)} H_\chi \subseteq \mathbb{R}^{\Omega_m}, \qquad H_\chi = \{\lambda : (\chi, \lambda) \ge 0\}. \tag{1.15}$$

A 1-PS $\lambda$ lies in $C_B$ if and only if $\lim_{\tau\to 0}\lambda(\tau)\cdot B$ exists. Put $H_\rho = \{\lambda : (\rho,\lambda)\geq 0\}$ and $H_\rho^\circ = \{\lambda : (\rho,\lambda) > 0\}$.

**Proposition 1.7** (Hilbert–Mumford criterion [Hos, §2.1])**.** *Let $B \in E_m$ and recall $C_B$ from (1.15). Then $B$ is $\rho$-semistable if and only if $C_B \subseteq H_\rho$, and $\rho$-stable if and only if $C_B\setminus\{0\} \subseteq H_\rho^\circ$.*

1.8. **Reachability and $U_m$.**

**Definition 1.8.** Let $B = (B', a) \in E_m$ (Definition 1.4). A slot $d$ is *reachable* in $B$ if there are slots $d_1, \dots, d_r = d$ ($r \geq 1$) and labels such that $a_{d_1}^{(p_0)} \neq 0$ and $B_{d_j\to d_{j+1}}^{(p_j)} \neq 0$ for $1 \leq j < r$. Let $R(B)$ be the set of reachable slots and $S(B) = \Omega_m \setminus R(B)$ the set of unreachable ones. Put

$$U_m = \{B \in E_m : R(B) = \Omega_m\}.$$

For $S \subseteq \Omega_m$ let $\lambda_S = -1_S$ be the 1-PS with $\lambda_S(\tau)_d = \tau^{-1}$ for $d \in S$ and 1 otherwise; $\lambda_\emptyset$ is the trivial 1-PS.

"$d$ is reachable" is a finite union of non-vanishing conditions, so $U_m$ is open; it is stable under $T_m \rtimes W_m$ (§1.6).

**Lemma 1.9.** *Let $B \in E_m$. Recall the sets $R(B)$, $S(B)$ and the 1-PS $\lambda_S$ of Definition 1.8, and the cone $C_B$ of (1.15).*

*(1) Every $\lambda \in C_B$ satisfies $\lambda_d \geq 0$ for all $d \in R(B)$.*
*(2) $\lambda_{S(B)} \in C_B$, and $(\rho, \lambda_{S(B)}) = -|S(B)|$.*

*Proof.* (1) Take a path $d_1, \dots, d_r = d$ as in Definition 1.8. By (1.13), $x_{d_1} \in \mathrm{wt}(B)$ and $x_{d_{j+1}} - x_{d_j} \in \mathrm{wt}(B)$, so $\lambda_{d_1} \geq 0$ and $\lambda_{d_{j+1}} \geq \lambda_{d_j}$; hence $\lambda_d \geq 0$.

(2) Put $S = S(B)$ and $R = R(B)$. A nonzero framing coordinate $a_d$ with $d \in S$, or a nonzero coordinate $B_{e\to d}$ with $e \in R$, $d \in S$, would make $d$ reachable; so these vanish. The remaining weights in $\mathrm{wt}(B)$ are $x_d$ with $d \in R$, $x_e - x_d$ with $d, e$ both in $R$ or both in $S$, and $x_e - x_d$ with $d \in S$, $e \in R$. Their pairings with $\lambda_S = -1_S$ are 0, 0 and 1, all $\geq 0$. Finally $(\rho, \lambda_S) = -|S|$ by (1.14). □

**Lemma 1.10.** *Let $B \in E_m$, and recall $\rho$-(semi)stability from Definition 1.6.*

*(1) The smallest subrepresentation of $B$ containing the space at $\infty$ is supported on $\{\infty\}\cup R(B)$. Hence $B \in U_m$ if and only if $B$ is cyclic, i.e. generated by the space at $\infty$.*
*(2) $U_m = E_m^{\rho\text{-}ss} = E_m^{\rho\text{-}s}$.*
*(3) $T_m$ acts freely on $U_m$. The quotient $U_m/T_m = E_m/\!/_\rho T_m$ is a smooth quasi-projective toric variety of dimension $\dim E_m - |l(m)|$ (if nonempty), and*

$$H^*_{T_m}(U_m) = H^*(U_m/T_m), \qquad H^*_{T_m}(U_m)^{W_m} = H^*\big((U_m/T_m)/W_m\big).$$

*Proof.* (1) All vertex spaces are one-dimensional, so a subrepresentation is a subset of vertices closed under nonzero arrows. The smallest one containing $\infty$ is $\{\infty\} \cup R(B)$.

(2) If $B \in U_m$, then every $0 \neq \lambda \in C_B$ has $\lambda_d \geq 0$ for all $d$ by Lemma 1.9(1), hence $(\rho, \lambda) = \sum_d \lambda_d > 0$. By Proposition 1.7, $B$ is $\rho$-stable. If $S(B) \neq \emptyset$, then Lemma 1.9(2) gives $\lambda_{S(B)} \in C_B$ with $(\rho, \lambda_{S(B)}) < 0$, so $C_B \not\subseteq H_\rho$ and $B$ is $\rho$-unstable.

(3) If $t\cdot B = B$, then $t_d = 1$ whenever $a_d \neq 0$, and $t_e = t_d$ whenever $B_{d\to e} \neq 0$; following paths from $\infty$ gives $t = 1$. By (2) and [Hos, Theorem 2.2], $U_m \to E_m/\!/_\rho T_m$ is a geometric quotient, smooth because the action is free. The torus $(\mathbb{C}^\times)^{\dim E_m}$ scaling all coordinates commutes with $T_m$ and preserves $U_m$, since reachability only depends on which coordinates are nonzero. If $U_m \neq \emptyset$, it contains the open orbit where all coordinates are nonzero, so $U_m/T_m$ is toric. The last assertions hold because the action is free and $W_m$ is finite. □

1.9. **The toric hyperKähler variety $Y_m$.** Choose an *orientation* $o$: for every unordered pair $\{d,e\}$ of distinct slots and every $1 \le p \le b_{de}$, declare one of $(d \to e, p)$, $(e \to d, p)$ to be *original*. Let $V_m^o$ be the framed quiver with vertex set $\Omega_m \sqcup \{\infty\}$, the original arrows and the framing arrows, and let

$$L_m^o = \bigoplus_{\text{original arrows}} \mathbb{C} \;\oplus\; F_m$$

be its representation space with dimension vector 1. The coordinate of the reverse $\bar\gamma$ of an original arrow $\gamma$ has the opposite weight, so it is naturally dual to the coordinate $B_\gamma$. Hence

$$T^*L_m^o = L_m^o \oplus (L_m^o)^\vee = E_m \oplus F_m^\vee, \qquad F_m^\vee = \bigoplus_d \left(\mathbb{C}^{w_{i(d)}}\right)^\vee,$$

where the coordinates $b_d = (b_d^{(p)})_p$ of $F_m^\vee$ are dual framing arrows $d \to \infty$, of weight $-x_d$. Following the recent literature we write $b$ for the dual framing; it should not be confused with the integers $b_{de}$ of (1.8) or the bilinear form $b(k,m)$ of (1.5). The moment map $\mu : E_m \oplus F_m^\vee \to \mathrm{Lie}(T_m)^\vee = \mathbb{C}^{\Omega_m}$, defined by $\langle \mu(s,p), \xi\rangle = \langle p, \xi \cdot s\rangle$ as in [Hik], is

$$\mu_d(B,b) = \sum_{\substack{\gamma \text{ original}\\ h(\gamma)=d}} B_\gamma B_{\bar\gamma} - \sum_{\substack{\gamma \text{ original}\\ t(\gamma)=d}} B_\gamma B_{\bar\gamma} + \langle b_d, a_d\rangle, \qquad \langle b_d, a_d\rangle = \sum_p b_d^{(p)} a_d^{(p)}.$$

**Definition 1.11.** $Y_m = \{(B,b) \in U_m \times F_m^\vee : \ \mu(B,b) = 0\}/T_m$, where $B = (B', a) \in U_m$ and $b \in F_m^\vee$.

*Remark* 1.12. (i) $Y_m$ is the quiver variety [Nak94] of $V_m^o$ with dimension vector $(1,\dots,1)$ and framing $w$, in its holomorphic description at determinant stability (cf. [Nak94, Theorem 3.2, Proposition 3.5]). The multipartition quiver has no edge loops, so this is within the setting of [Nak94]. Indeed, by the argument of Lemma 1.10(1), cyclicity in the doubled quiver means reachability using the framing arrows $\infty \to d$ and the arrows between slots in both directions; that is exactly the condition $B \in U_m$, and the dual framing $b$ plays no role.

(ii) $T_m$ acts freely on $U_m \times F_m^\vee$ (Lemma 1.10(3)), so $\mu$ is a submersion there and $Y_m$ is smooth of dimension

$$\dim Y_m = 2g_m, \qquad g_m = \sum_{\{d,e\}} b_{de} + w(m) - |l(m)|.$$

Here $g_m$ is the number of edges minus the number of slots of the underlying graph of $\Gamma_m$, i.e. its first Betti number when that graph is connected.

(iii) Changing the orientation changes the sign of some terms of $\mu$. Changing the sign of the corresponding reverse coordinates is a $T_m$-equivariant isomorphism, so $Y_m$ does not depend on $o$ up to isomorphism. Composing with such sign changes, $W_m$ acts on $Y_m$.

(iv) *Relation with Hausel–Sturmfels* [HS]. Let $A$ be the integer matrix with $|l(m)|$ rows whose columns are the $T_m$-weights (1.13) of the coordinates of $L_m^o$, one column for each original arrow and each framing arrow. This is the boundary map [HS, (40)] of the quiver $V_m^o$, with $v_0 = \infty$; it is unimodular [HS, Lemma 8.1], and $T_m$ plays the role of the torus $\mathbb{T}_{\mathbb{C}}^d$ of [HS]. The character $\rho = \det$ of (1.14) is the vector $\theta = (1,\dots,1)$ of [HS]. A hyperplane spanned by columns of $A$ is spanned by the arrows of a forest with two components; if $S \subseteq \Omega_m$ is the component not containing $\infty$, the hyperplane is $\{y : \sum_{d\in S} y_d = 0\}$. Since $\sum_{d\in S}\theta_d = |S| \neq 0$, $\theta$ is generic in the sense of [HS, §4]. In this notation:

- The coordinates $z_i, w_i$ of [HS, §4] are the coordinates of $L_m^o$ and their duals, so the Lawrence toric variety of [HS, Definition 4.1, Theorem 1.1(2)] is the GIT quotient of $T^*L_m^o = E_m \oplus F_m^\vee$ without the moment map equation:

$$X_m^{\mathrm{Law}} := X(A^{\pm}, \theta) = (E_m \oplus F_m^\vee)/\!/_\rho\, T_m.$$

- $Y_m$ is the toric quiver variety $Y(V_m^o,\theta) = Y(A,\theta)$ of [HS, §8]: the subvariety of $X_m^{\mathrm{Law}}$ cut out by the components of the complex moment map [HS, (29), Definition 6.1], biholomorphic to the hyperKähler quotient [HS, Theorem 6.3]. (In [HS, §6] the Gale dual matrix is assumed to have no zero row, i.e. the underlying graph has no bridge, so that the ideal (29) is prime; in general we take Definition 1.11 as the definition.)
- $X_m^{\mathrm{Law}}$ is a vector bundle over the toric manifold $U_m/T_m$, whose fiber consists of the linear maps *to* the framing vector spaces. Indeed, the coordinates $b_d$ of $F_m^\vee$ are arrows $d \to \infty$, which do not help to reach slots from $\infty$; so Lemma 1.9(1) still holds for the cone of allowable 1-PSs of $(B,b)$. Their weights $-x_d$ pair with $\lambda_{S(B)}$ to 0 or 1, so Lemma 1.9(2) also holds. The proof of Lemma 1.10(2) then shows that the semistable locus is $U_m \times F_m^\vee$, on which $T_m$ acts freely. Therefore

$$X_m^{\mathrm{Law}} = U_m \times^{T_m} F_m^\vee = \bigoplus_{d\in\Omega_m} \mathrm{Hom}\left(\mathcal{L}_d,\, \mathbb{C}^{w_{i(d)}}\right),$$

  where $\mathcal{L}_d$ is the line bundle on $U_m/T_m$ associated with the character $t \mapsto t_d$ (the tautological line at the slot $d$), and $\mathbb{C}^{w_{i(d)}}$ is a trivial bundle. The zero section is $U_m/T_m$, and $Y_m \subset X_m^{\mathrm{Law}}$ is the closed subvariety $\mu = 0$. For a single slot with $w$ framing arrows this is the example in [HS, §1]: a rank $w$ bundle over $\mathbb{P}^{w-1}$ containing $Y_m = T^*\mathbb{P}^{w-1}$.
- By [HS, Theorem 1.1] and the remark after its proof, $Y_m$, $X_m^{\mathrm{Law}}$ and their common core are deformation retracts of one another, and their cohomology (with $\mathbb{Z}$ coefficients, as $A$ is unimodular) is $\mathbb{Z}[x_1,\dots,x_N]/(\mathrm{Circ}(A)+M^*(A))$, where $N$ is the number of columns of $A$ and there is one variable for each column. By [HS, Theorem 8.3], $M^*(A)$ is generated by the monomials of the cuts of the underlying graph. In the presentation [HS, (37)], which eliminates $\mathrm{Circ}(A)$, each variable is replaced by the weight of its arrow, a linear form in the $x_d$; the monomial of the cut separating $S$ from $(\Omega_m \setminus S)\cup\{\infty\}$ then becomes $\pm q_S$ (Definition 1.5). Hence

$$H^*(Y_m) \cong H^*(X_m^{\mathrm{Law}}) \cong P_m/J_m,$$

  and $J_m$ is the image of the cocircuit ideal; compare [Hik, Section 5.1]. Combined with the vector bundle structure, this gives a second proof of Proposition 2.1(a), the map $P_m \to H^*(U_m/T_m)$ sending $x_d$ to $c_1(\mathcal{L}_d)$. By [HS, Corollary 8.4], the Euler characteristic of $Y_m$ is the number of spanning trees of the underlying graph; for $m_b$ in Example 1.13 it is 4. (If some component of the graph misses $\infty$, then $U_m = \emptyset$ and $J_m = P_m$, so all these spaces are empty.)

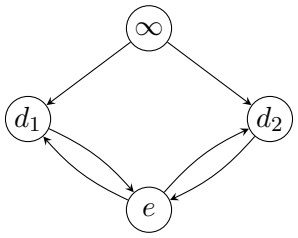
Apart from the alternative argument for the finite-dimensionality of $P_m/J_m$ in §1.5, none of this is used in what follows.

(v) Applying Crawley-Boevey–Van den Bergh [CBVdB] to $V_m^o$ with dimension vector $(1,\dots,1)$, and using (iv) and purity, the number $a_m(q)$ of absolutely indecomposable representations of $V_m^o$ of this dimension over $\mathbb{F}_q$ is $a_m(q) = q^{-g_m}\#Y_m(\mathbb{F}_q) = q^{g_m}\,\mathrm{Hilb}(P_m/J_m; q^{-1})$. By Kac, $a_m$ does not depend on $o$. This is also not used below.

**Example 1.13.** Let $Q$ be the quiver $1 \to 2$ with a loop added at each vertex, so $Q'$ is $1 \to 2$, $c_{12} = c_{21} = 1$ and $c_{11} = c_{22} = 0$ in (1.1). Take $v = (2,1)$ and $w = (1,0)$. Then $C(v) = 2$, and $\mathcal{M}(v)$ consists of

$$m_a:\ \lambda^1 = (2),\ \lambda^2 = (1), \qquad m_b:\ \lambda^1 = (1,1),\ \lambda^2 = (1).$$

For $m_b$ there are slots $d_1, d_2$ of type $(1,1)$ and $e$ of type $(2,1)$, with $b_{d_1e} = b_{d_2e} = 1$ and $b_{d_1d_2} = c_{11} = 0$; $W_{m_b} = \mathfrak{S}_2$ swaps $d_1, d_2$. The multipartition quiver is




so $\dim E_{m_b} = 6$ and $U_{m_b}/T_{m_b}$ is a smooth toric threefold. Its underlying graph is a 4-cycle, $g_{m_b} = 1$, and $Y_{m_b}$ is the $A_3$-surface. Writing $a = x_{d_1}$, $b = x_{d_2}$, $c = x_e$, the cut polynomials of Definition 1.5 are

$$a(a-c),\ b(b-c),\ (c-a)(c-b),\ a(c-b),\ b(c-a),\ ab,\ ab(a-c)(b-c),$$

for $S = \{d_1\}, \{d_2\}, \{e\}, \{d_1, e\}, \{d_2, e\}, \{d_1, d_2, e\}, \{d_1, d_2\}$ respectively. They generate $J_{m_b} = (a, b, c)^2$. Hence $P_{m_b}/J_{m_b}$ has basis $1, a, b, c$, with Hilbert series $1 + 3q$, and $h_{m_b} = 1 + 2q$ (invariants 1, $a + b$, $c$).

For $m_a$ there are slots $d$ of type $(1, 2)$ and $e$ of type $(2, 1)$, with $b_{de} = \min(2, 1) = 1$; the multipartition quiver is $\infty \to d \rightleftarrows e$. Then $J_{m_a} = (x_d, x_e)$ and $h_{m_a} = 1$.

## 2. Proposition 6.2 via Kirwan's equivariant perfection

Recall the multipartition space $E_m$ (Definition 1.4) and its open subset $U_m$ (Definition 1.8), the torus $T_m$ and $P_m = H^*_{T_m}(\mathrm{pt})$ (§1.5), the cut ideal $J_m$ and the polynomial $h_m$ (Definition 1.5), and the notation $(q)_m$, $w(k)$, $b(k, m)$ of Definition 1.2.

**Proposition 2.1** ([Hik, Proposition 6.2 and (6.2)])**.** *For every multipartition $m$:*

*(a) $H^*_{T_m}(U_m)$ is concentrated in even degrees, and restriction $P_m = H^*_{T_m}(E_m) \to H^*_{T_m}(U_m)$ is surjective with kernel $J_m$. In particular $H^*(U_m/T_m) \cong P_m/J_m$, and $h_m$ is the Poincaré polynomial of $(U_m/T_m)/W_m$.*

*(b) In $\mathbb{Z}[[q]]$,*

$$\frac{1}{(q)_m} = \sum_{k \le m} q^{\,w(k)+b(k,m-k)}\ \frac{h_{m-k}(q)}{(q)_k}. \tag{6.1}$$

The term $k = \emptyset$ in (6.1) is $h_m$, so (6.1) determines $h_m$ recursively. The proof below is Kirwan's argument [Kir], applied to Hesselink's stratification of $E_m$ for the torus $T_m$ and the character $\rho = \det$ of (1.14), and made $W_m$-equivariant.

2.1. **Recollection: Hesselink's stratification.** We recall [Hos, §§2.2–2.3] for a torus $T$ acting linearly on a complex vector space $V$, with a character $\rho$ and a norm $\|\cdot\|$ on 1-PSs as in §1.7. For $v \in V$ let $C_v$ be the cone of allowable 1-PSs, defined as in (1.15), so that $\lambda \in C_v$ if and only if $\lim_{\tau\to 0} \lambda(\tau) \cdot v$ exists. For a $\rho$-unstable $v$, [Hos, Definition 2.12] puts

$$M^\rho(v) = \inf\left\{\frac{(\rho, \lambda)}{\|\lambda\|} :\ 0 \ne \lambda \in C_v\right\} < 0.$$

A 1-PS $\lambda$ is *$\rho$-adapted* to $v$ if it attains this infimum, and $\wedge^\rho(v)$ denotes the set of indivisible $\rho$-adapted 1-PSs; for a torus it has exactly one element [Hos, Lemma 2.13]. For a 1-PS $\lambda$ the *Hesselink stratum* (Hoskins' blade, which equals the stratum $S_{[\lambda]}$ since $T$ is abelian) and the *limit set* are [Hos, (5), (6)]

$$S_\lambda = \{v\ \rho\text{-unstable} :\ \lambda \in \wedge^\rho(v)\}, \qquad Z_\lambda = \{v \in S_\lambda :\ \lambda(\tau) \cdot v = v \text{ for all } \tau\},$$

and $S_0 := V^{\rho\text{-}ss}$ for the trivial 1-PS. Let $V^\lambda$ be the fixed locus of $\lambda$, $V^\lambda_+$ the set of points $v$ for which $\lim_{\tau\to 0}\lambda(\tau)\cdot v$ exists, and $p_\lambda : V^\lambda_+ \to V^\lambda$ the limit map. These are linear subspaces and a linear projection. By [Hos, Proposition 2.18, Lemma 2.19],

$$S_\lambda = p_\lambda^{-1}(Z_\lambda),$$

and $Z_\lambda$ is the semistable locus for $T$ acting on $V^\lambda$ with respect to the character $\rho_\lambda = \|\lambda\|^2\rho - (\rho,\lambda)\lambda^*$, where $\lambda^*$ is the character dual to $\lambda$ under the inner product. In particular $Z_\lambda$ is open in $V^\lambda$ and $S_\lambda$ is open in $V^\lambda_+$; the normal space of $S_\lambda$ is $V^\lambda_- = \bigoplus_{(\chi,\lambda)<0} V_\chi$. By [Hos, Theorem 2.16, Remark 2.17], $V = \bigsqcup_\lambda S_\lambda$ is a decomposition into finitely many $T$-invariant locally closed subvarieties, *Hesselink's stratification*, and $\partial S_\lambda \cap S_{\lambda'} \neq \emptyset$ only if $\lambda < \lambda'$, where $\lambda < \lambda'$ means $(\rho,\lambda)/\|\lambda\| > (\rho,\lambda')/\|\lambda'\|$. By [Hos, Theorems 4.12 and 4.18] it coincides with Kirwan's stratification and with the Morse stratification of the norm square of the shifted moment map.

*Kirwan's equivariant perfection* [Kir]: if the Euler class of each $V^\lambda_-$ is a non-zero-divisor in $H^*_T(S_\lambda)$ (the Atiyah–Bott criterion [AB]), then the Thom–Gysin sequences split, the restriction $H^*_T(V) \to H^*_T(V^{\rho\text{-}ss})$ is surjective, and

$$\operatorname{Hilb} H^*_T(V) = \operatorname{Hilb} H^*_T(V^{\rho\text{-}ss}) + \sum_{\lambda\neq 0} q^{\dim V^\lambda_-} \operatorname{Hilb} H^*_T(S_\lambda).$$

Below we verify everything we need directly for $V = E_m$.

### 2.2. The adapted 1-PS of a point of $E_m$.

**Lemma 2.2.** *Let $B \in E_m$. Recall the sets $R(B)$, $S(B)$ of reachable and unreachable slots and the 1-PS $\lambda_S = -1_S$ from Definition 1.8, and the cone $C_B$ of allowable 1-PSs from (1.15). If $S(B) \neq \emptyset$, then $B$ is $\rho$-unstable,*

$$M^\rho(B) = -\sqrt{|S(B)|}, \qquad \wedge^\rho(B) = \{\lambda_{S(B)}\}.$$

*Proof.* Put $S = S(B)$ and $R = R(B)$; $B$ is $\rho$-unstable by Lemma 1.10(2). Let $0 \neq \lambda \in C_B$. By Lemma 1.9(1), $\lambda_d \geq 0$ for $d \in R$, so by Cauchy–Schwarz

$$(\rho,\lambda) = \sum_{d\in R}\lambda_d + \sum_{d\in S}\lambda_d \;\geq\; \sum_{d\in S}\lambda_d \;\geq\; -\sqrt{|S|}\Big(\sum_{d\in S}\lambda_d^2\Big)^{1/2} \;\geq\; -\sqrt{|S|}\,\|\lambda\|.$$

By Lemma 1.9(2), $\lambda_S \in C_B$ and $(\rho,\lambda_S)/\|\lambda_S\| = -|S|/\sqrt{|S|} = -\sqrt{|S|}$. Hence $M^\rho(B) = -\sqrt{|S|}$, attained by $\lambda_S$. Equality throughout forces $\lambda_d = 0$ for $d \in R$ and $\lambda|_S$ to be a negative multiple of $1_S$, i.e. $\lambda \in \mathbb{R}_{>0}\lambda_S$. Since $\lambda_S$ is indivisible, $\wedge^\rho(B) = \{\lambda_S\}$. □

**Corollary 2.3.** *Hesselink's stratification of $E_m$ (§2.1) is*

$$E_m = \bigsqcup_{S\subseteq\Omega_m} S_{\lambda_S}, \qquad S_{\lambda_S} = \{B \in E_m : S(B) = S\},$$

*with $S_{\lambda_\emptyset} = S_0 = U_m$; all other 1-PSs index empty strata. The ordering is $\lambda_S < \lambda_{S'}$ if and only if $|S| < |S'|$. For $0 \leq r \leq |l(m)|$ put*

$$U_{m,r} = \{B \in E_m :\ |S(B)| \leq r\} = \bigsqcup_{|S|\leq r} S_{\lambda_S}.$$

*Then $U_{m,0} = U_m$, $U_{m,|l(m)|} = E_m$, $U_{m,r} \setminus U_{m,r-1} = \bigsqcup_{|S|=r} S_{\lambda_S}$, and each $U_{m,r}$ is open in $E_m$.*

*Proof.* The description of the strata follows from Lemma 2.2 and Lemma 1.10(2), and the ordering from $M^\rho = -\sqrt{|S|}$ on $S_{\lambda_S}$. The set $U_{m,r}$ is the union over $A \subseteq \Omega_m$ with $|A| = |l(m)| - r$ of the open sets $\{B : A \subseteq R(B)\}$. □

2.3. **The Hesselink strata of $E_m$.** Fix a nonempty $S \subseteq \Omega_m$, and put $R = \Omega_m \setminus S$, $k = \kappa(S)$ (the type of $S$, §1.3) and $m' = m - k$. Decompose

$$E_m = E_{RR} \oplus E_{SS} \oplus E_{SR} \oplus E_{RS} \oplus F_R \oplus F_S,$$

where $E_{XY}$ collects the coordinates $B^{(p)}_{d\to e}$ with $d \in X$, $e \in Y$, and $F_X$ the framing coordinates $a^{(p)}_d$ with $d \in X$. Via the identification (1.10), $R \cong \Omega_{m'}$, we identify $E_{RR} \oplus F_R = E_{m'}$ and $T_m = T_{m'} \times T_S$, where $T_S = (\mathbb{C}^\times)^S$. By (1.13), the $\lambda_S$-weights of the coordinates are $-1$ on $E_{RS} \oplus F_S$, $+1$ on $E_{SR}$ and 0 otherwise. Hence, in the notation of §2.1,

$$E_m^{\lambda_S} = E_{RR} \oplus E_{SS} \oplus F_R, \qquad (E_m)_+^{\lambda_S} = E_m^{\lambda_S} \oplus E_{SR}, \qquad N_S := (E_m)_-^{\lambda_S} = E_{RS} \oplus F_S, \tag{2.1}$$

and $p_{\lambda_S}$ kills $E_{SR}$.

**Lemma 2.4.** *In this notation, recall $U_{m,r}$ from Corollary 2.3 and $q_S$ from Definition 1.5.*

*(a) The limit set is $Z_{\lambda_S} = U_{m'} \times E_{SS} \subseteq E_m^{\lambda_S}$, and the Hesselink stratum is $S_{\lambda_S} = p_{\lambda_S}^{-1}(Z_{\lambda_S}) = U_{m'} \times E_{SS} \times E_{SR}$.*

*(b) $S_{\lambda_S} = (E_m)_+^{\lambda_S} \cap U_{m,|S|}$. In particular $S_{\lambda_S}$ is smooth and closed in $U_{m,|S|}$.*

*(c) The normal bundle of $S_{\lambda_S}$ in $U_{m,|S|}$ is the trivial bundle with fiber $N_S$. Its rank is $c_S = w(k) + b(k, m')$, and its equivariant Euler class is $e_S = q_S$.*

*(d) $H^*_{T_m}(S_{\lambda_S}) \cong H^*_{T_{m'}}(U_{m'}) \otimes \mathbb{C}[x_d : d \in S]$, compatibly with restriction from $P_m = P_{m'} \otimes \mathbb{C}[x_d : d \in S]$.*

*Proof.* (a) By Corollary 2.3, $S_{\lambda_S} = \{B : S(B) = S\}$. If $S(B) = S$, then $\lambda_S \in C_B$ by Lemma 1.9(2), i.e. $B \in (E_m)_+^{\lambda_S}$. For $B \in (E_m)_+^{\lambda_S}$ the coordinates in $E_{RS} \oplus F_S$ vanish, so a path from $\infty$ never enters $S$; hence $S \subseteq S(B)$, and reachability of the slots of $R$ only involves the coordinates in $E_{RR} \oplus F_R = E_{m'}$. Thus $S(B) = S$ if and only if $B|_{E_{m'}} \in U_{m'}$, with no condition on $E_{SS}$ and $E_{SR}$. This gives $S_{\lambda_S}$, and $Z_{\lambda_S} = S_{\lambda_S} \cap E_m^{\lambda_S} = U_{m'} \times E_{SS}$.

(b) If $B \in (E_m)_+^{\lambda_S} \cap U_{m,|S|}$, then $S \subseteq S(B)$ by the proof of (a) and $|S(B)| \le |S|$, so $S(B) = S$. The converse follows from (a). Since $(E_m)_+^{\lambda_S}$ is closed, $S_{\lambda_S}$ is closed in $U_{m,|S|}$; by (a) it is open in the linear space $(E_m)_+^{\lambda_S}$, hence smooth.

(c) Since $S_{\lambda_S}$ is open in $(E_m)_+^{\lambda_S}$, its normal bundle is $S_{\lambda_S} \times (E_m/(E_m)_+^{\lambda_S}) = S_{\lambda_S} \times N_S$. By (1.13), the weights of $N_S$ are $x_d$, with multiplicity $w_{i(d)}$, for the framing coordinates at $d \in S$, and $x_d - x_e$, with multiplicity $b_{ed} = b_{de}$, for the coordinates $B_{e\to d}$ with $e \in R$, $d \in S$. So the rank is $w(k) + b(k, m')$ by (1.3) and (1.11), and the Euler class is $\prod_{d\in S} x_d^{w_{i(d)}} \prod_{d\in S, e\in R} (x_d - x_e)^{b_{de}} = q_S$.

(d) The linear contraction of $E_{SS} \times E_{SR}$ retracts $S_{\lambda_S}$ equivariantly onto $U_{m'}$, on which $T_S$ acts trivially; so $H^*_{T_{m'} \times T_S}(U_{m'}) = H^*_{T_{m'}}(U_{m'}) \otimes H^*(BT_S)$. □

*Remark* 2.5. This agrees with [Hos, Proposition 2.18]: $\lambda_S^* = -1_S$, so $\rho_{\lambda_S} = |S|\, 1_{\Omega_m} - |S|\, 1_S = |S|\, 1_R$. A point of $E_m^{\lambda_S} = E_{m'} \oplus E_{SS}$ is semistable for this character exactly when its $E_{m'}$-component lies in $U_{m'}$, by Lemma 1.10(2) for $m'$, since $1_R$ is trivial on $T_S$.

2.4. **The Atiyah–Bott criterion.**

**Lemma 2.6.** *Let $S \neq \emptyset$. Multiplication by the Euler class $e_S = q_S$ of Lemma 2.4(c) on $H^*_{T_m}(S_{\lambda_S})$ is injective.*

*Proof.* By Lemma 2.4(d), $H^*_{T_m}(S_{\lambda_S}) = A[x_d : d \in S]$ with $A = H^*_{T_{m'}}(U_{m'})$, an arbitrary commutative ring (possibly with nilpotents) containing the $x_e$, $e \in R$. Regard $q_S$ (Definition 1.5) as a polynomial in the variables $x_S$ over $A$. Expanding $(x_d - x_e)^{b_{de}}$ shows that its part of top total degree in $x_S$ is the single monomial $\prod_{d\in S} x_d^{w_{i(d)} + \sum_{e\in R} b_{de}}$, with coefficient 1. If $f \neq 0$, the top-degree part of $q_S f$ is this monomial times the top-degree part of $f$, which is nonzero. □

*Remark* 2.7. This is the Atiyah–Bott criterion [AB]. The 1-PS $\lambda_S^{-1}$ acts trivially on the retract $U_{m'} \times \{0\}$ of $S_{\lambda_S}$ and with weight $+1$ on every coordinate of $N_S$. Restricting to it sends every factor of $q_S$ to the generator $u \in H^2(B\mathbb{C}^\times)$, so $q_S \mapsto u^{c_S}$; this is exactly the leading monomial used above.

2.5. **Equivariant perfection.**

**Proposition 2.8.** *Recall $U_{m,r}$ from Corollary 2.3 and $c_S$ from Lemma 2.4(c). For $1 \le r \le |l(m)|$ there is a short exact sequence*

$$0 \to \bigoplus_{|S|=r} H_{T_m}^{*-2c_S}(S_{\lambda_S}) \xrightarrow{\sum_S i_{S*}} H_{T_m}^*(U_{m,r}) \to H_{T_m}^*(U_{m,r-1}) \to 0,$$

*where $i_S : S_{\lambda_S} \hookrightarrow U_{m,r}$ is the inclusion.*

*Proof.* By Corollary 2.3 and Lemma 2.4(b), the union of the strata $S_{\lambda_S}$ with $|S| = r$ is a closed smooth subvariety of $U_{m,r}$, with complement $U_{m,r-1}$. So there is the equivariant Thom–Gysin long exact sequence

$$\cdots \to \bigoplus_{|S|=r} H_{T_m}^{k-2c_S}(S_{\lambda_S}) \to H_{T_m}^k(U_{m,r}) \to H_{T_m}^k(U_{m,r-1}) \xrightarrow{\delta} \bigoplus_{|S|=r} H_{T_m}^{k+1-2c_S}(S_{\lambda_S}) \to H_{T_m}^{k+1}(U_{m,r}) \to \cdots.$$

By the self-intersection formula $i_S^* i_{S*} = e_S \cup (-) = q_S \cup (-)$, which is injective by Lemma 2.6. For $S' \neq S$ with $|S'| = |S|$ the strata are disjoint, so $i_{S'}^* i_{S*} = 0$. Hence $\sum_S i_{S*}$ is injective in every degree, the connecting maps $\delta$ vanish, and the long exact sequence splits into short exact sequences. □

In particular, restriction $H_{T_m}^*(E_m) \to H_{T_m}^*(U_m)$ is surjective: this is Kirwan surjectivity for the toric quotient $U_m/T_m$. Odd cohomology vanishes at every stage, because $\delta = 0$ and $H_{T_m}^{\mathrm{odd}}(E_m) = 0$.

2.6. **Proof of Proposition 2.1(a).** Recall the cut polynomials $q_S$ and the cut ideal $J_m$ from Definition 1.5. We use induction on the number of slots $|l(m)|$; for $m = \emptyset$ there is nothing to prove. Assume Proposition 2.1(a) for all multipartitions with fewer slots than $m$. For $0 \le r \le |l(m)|$ let $J_{m,r} \subseteq P_m$ be the ideal generated by the $q_S$ with $|S| > r$, so that $J_{m,0} = J_m$. We show by descending induction on $r$ that restriction $P_m \to H_{T_m}^*(U_{m,r})$ is surjective with kernel $J_{m,r}$. For $r = |l(m)|$ we have $U_{m,r} = E_m$ and $J_{m,r} = 0$.

Assume the claim for $r$. By Proposition 2.8, $H_{T_m}^*(U_{m,r-1})$ is the quotient of $H_{T_m}^*(U_{m,r}) = P_m/J_{m,r}$ by the images of the Gysin maps $i_{S*}$, $|S| = r$. Since $S \neq \emptyset$, $m' = m - \kappa(S)$ has fewer slots, so by induction and Lemma 2.4(d), $P_m = P_{m'} \otimes \mathbb{C}[x_S] \to H_{T_m}^*(S_{\lambda_S})$ is surjective. Thus every class on $S_{\lambda_S}$ is $i_S^* f$ for some $f \in P_m$, and by the projection formula $i_{S*}(i_S^* f) = f \cup i_{S*}(1)$. Moreover $i_{S*}(1) = [S_{\lambda_S}]$ is the restriction to $U_{m,r}$ of the equivariant class of the linear subspace $(E_m)_+^{\lambda_S} \subseteq E_m$, which is the Euler class $e(N_S) = q_S$; here we use $S_{\lambda_S} = (E_m)_+^{\lambda_S} \cap U_{m,r}$ (Lemma 2.4(b)). Hence the image of the Gysin maps is the ideal generated by the $q_S$ with $|S| = r$, and $H_{T_m}^*(U_{m,r-1}) = P_m/(J_{m,r} + (q_S : |S| = r)) = P_m/J_{m,r-1}$. For $r = 0$ this gives $H_{T_m}^*(U_m) = P_m/J_m$. Evenness was shown after Proposition 2.8, and the statements about $U_m/T_m$ follow from Lemma 1.10(3). □

2.7. **Proof of Proposition 2.1(b): taking $W_m$-invariants.** Recall the action of $W_m$ on $E_m$ and on equivariant cohomology from §1.6, and the types of subsets and their stabilizers from §1.3. Since $S(\sigma B) = \sigma S(B)$, $W_m$ preserves each $U_{m,r}$ and permutes the strata: $\sigma(S_{\lambda_S}) = S_{\lambda_{\sigma S}}$. Restriction maps commute with $W_m$. The Gysin maps do as well, because $\sigma$ induces complex-linear isomorphisms $N_S \to N_{\sigma S}$ (permutations of coordinates), so it preserves the Thom classes defined by the complex orientations, without signs. Hence the sequences of Proposition 2.8 are sequences of $W_m$-modules, and they stay exact after taking invariants (§1.6(a)).

Fix $r \geq 1$. For each $k \leq m$ with $|l(k)| = r$ choose $S_k \subseteq \Omega_m$ of type $k$. The subsets of type $k$ form the orbit $W_m \cdot S_k$, with stabilizer $W_k \times W_{m-k}$, and $c_S = w(k) + b(k, m-k)$ is constant on it. For a $W_m$-module $\bigoplus_{S \in W_m S_k} M_S$ permuted transitively, projection to the $S_k$-summand identifies the invariants with $M_{S_k}^{W_k \times W_{m-k}}$; in particular no binomial coefficients appear. By Lemma 2.4(d) and part (a) for $m' = m - k$, compatibly with the actions of $W_{m'}$ and $W_k$,

$$H^*_{T_m}(S_{\lambda_{S_k}})^{W_k \times W_{m'}} \cong (P_{m'}/J_{m'})^{W_{m'}} \otimes \mathbb{C}[x_d : d \in S_k]^{W_k},$$

whose Hilbert series is $h_{m'}(q)/(q)_k$ by Definition 1.5 and §1.6(c). Therefore

$$\operatorname{Hilb}\left(H^*_{T_m}(U_{m,r})^{W_m}\right) - \operatorname{Hilb}\left(H^*_{T_m}(U_{m,r-1})^{W_m}\right) = \sum_{\substack{k \leq m \\ |l(k)|=r}} q^{\,w(k)+b(k,m-k)} \frac{h_{m-k}(q)}{(q)_k}.$$

Summing over $r = 1, \ldots, |l(m)|$ telescopes. At $r = |l(m)|$ we have $U_{m,r} = E_m$ and $\operatorname{Hilb}(P_m^{W_m}) = 1/(q)_m$ by §1.6(c); at $r = 0$ we have $\operatorname{Hilb}(H^*_{T_m}(U_m)^{W_m}) = h_m$ by part (a). This gives (6.1). □

2.8. **Dictionary and an example.** The following table summarizes the correspondence with [Hos, §2] and [Kir].

| Hoskins [Hos, §2] | multipartition space $E_m$, character $\rho = \det$ |
|---|---|
| $\rho$-semistable locus $V^{\rho\text{-}ss} = S_0$ | $U_m$ (Lemma 1.10(2)) |
| cone of allowable 1-PSs $C_v$ | $C_B$; $\lambda_d \geq 0$ on $R(B)$ (Lemma 1.9) |
| $M^\rho(v)$, $\wedge^\rho(v)$ | $-\sqrt{\lvert S(B)\rvert}$, $\{\lambda_{S(B)}\}$ (Lemma 2.2) |
| ordering $\lambda < \lambda'$ | $\lvert S\rvert < \lvert S'\rvert$ (Corollary 2.3) |
| $V^\lambda$, $V^\lambda_+$ | $E_{RR} \oplus E_{SS} \oplus F_R$, $E_m^{\lambda_S} \oplus E_{SR}$ (2.1) |
| limit set $Z_\lambda$ | $U_{m'} \times E_{SS}$ (Lemma 2.4(a)) |
| Hesselink stratum $S_\lambda = p_\lambda^{-1}(Z_\lambda)$ | $\{S(B) = S\} = U_{m'} \times E_{SS} \times E_{SR}$ |
| normal space $V^\lambda_-$, codimension | $N_S = E_{RS} \oplus F_S$, $w(k) + b(k, m')$ |
| Euler class of $V^\lambda_-$ | cut polynomial $q_S$ (Lemma 2.4(c)) |
| Atiyah–Bott criterion | leading monomial of $q_S$ (Lemma 2.6) |
| Kirwan's formula | (6.1), in $W_m$-invariants |

**Example 2.9.** For $m_b$ in Example 1.13, $(q)_{m_b} = (1-q)^2(1-q^2)$, and the six $k \leq m_b$, written as $(k_{1,1}, k_{2,1})$, contribute to (6.1):

| $k$ | $w(k)+b(k,m_b-k)$ | $h_{m_b-k}$ | term |
|---|---|---|---|
| $(0,0)$ | $0$ | $1+2q$ | $1+2q$ |
| $(1,0)$ | $1+1$ | $1$ | $q^2/(1-q)$ |
| $(2,0)$ | $2+2$ | $0$ | $0$ (slot $e$ alone has no framing, $U=\emptyset$) |
| $(0,1)$ | $0+2$ | $1$ | $q^2/(1-q)$ |
| $(1,1)$ | $1+1$ | $1$ | $q^2/(1-q)^2$ |
| $(2,1)$ | $2+0$ | $1$ | $q^2/\big((1-q)^2(1-q^2)\big)$ |

Multiplying by $(1-q)^2(1-q^2)$, the right side of (6.1) becomes $(1+2q)(1-2q+2q^3-q^4)+2q^2(1-q)(1-q^2)+q^2(1-q^2)+q^2=1$, as it should.

## 3. Hausel's formula and Corollary 6.3

3.1. **Hausel's formula.** In this subsection $Q$ is an arbitrary finite quiver; loops and multiple arrows are allowed. For dimension vectors $v,w$ let $G=\prod_i GL(V_i)$, $N=\bigoplus_{a\in Q_1}\mathrm{Hom}(V_{t(a)},V_{h(a)})\oplus\bigoplus_i\mathrm{Hom}(V_i,W_i)$, let $\mu: T^*N\to\mathfrak{g}^\vee\cong\mathfrak{g}$ be the moment map, and put

$$\mathfrak{M}_\eta(v,w)=\mu^{-1}\big((\mathrm{id}_{V_i})_i\big)/\!/G.$$

When nonempty, this is smooth of dimension $2\delta(v,w)$, where

$$\delta(v,w)=\sum_{a\in Q_1}v_{t(a)}v_{h(a)}-\sum_i v_i^2+w\cdot v.$$

Its odd cohomology vanishes; write $P_{\mathfrak{M}_\eta(v,w)}(q)$ for its Poincaré polynomial in the half-degree variable $q$. For partitions $\mu,\nu$ let $\langle\mu,\nu\rangle=\sum_{a,b}\min(\mu_a,\nu_b)$, let $(1^{w_i})$ be the partition with $w_i$ parts equal to 1, and for a $Q_0$-tuple of partitions $\lambda$ put $|\lambda|=(|\lambda^i|)_i$ and

$$E_Q(\lambda)=\sum_{a\in Q_1}\langle\lambda^{t(a)},\lambda^{h(a)}\rangle-\sum_i\langle\lambda^i,\lambda^i\rangle.$$

**Theorem 3.1** (Hausel [Hau]; [Hik, (6.9)])**.** *In* $\mathbb{C}(q)[[z_i:i\in Q_0]]$,

$$\sum_v q^{-\delta(v,w)}P_{\mathfrak{M}_\eta(v,w)}(q)\,z^v=\frac{\displaystyle\sum_\lambda z^{|\lambda|}\,\frac{q^{-E_Q(\lambda)-\sum_i\langle\lambda^i,(1^{w_i})\rangle}}{(q)_{m(\lambda)}}}{\displaystyle\sum_\lambda z^{|\lambda|}\,\frac{q^{-E_Q(\lambda)}}{(q)_{m(\lambda)}}},\tag{3.1}$$

*where* $\lambda$ *runs over all* $Q_0$*-tuples of partitions, including the empty one, and* $(q)_{m(\lambda)}$ *is as in* (1.4).

We do not discuss Hausel's proof, which uses an arithmetic Fourier transform. Remark 3.2 explains the alternative point-count proof of [Nak04], and Remark 3.3 relates $\mathfrak{M}_\eta(v,w)$ to the quiver variety at determinant stability.

*Remark* 3.2 (The point-count proof, after [Nak04])*.* We explain the right side of (3.1) following the unpublished note [Nak04, §1], which obtains (3.1) from the Kac–Stanley formula [Kac, Hua], Crawley-Boevey's trick [CB] and a theorem of Crawley-Boevey and Van den Bergh [CBVdB]. Work over a finite field $\mathbb{F}_u$ and put

$$R_Q(V)=\bigoplus_{a\in Q_1}\mathrm{Hom}(V_{t(a)},V_{h(a)}),\qquad R_Q(V,W)=R_Q(V)\oplus\bigoplus_i\mathrm{Hom}(W_i,V_i),$$

with $G_V=\prod_i GL(V_i)$ acting trivially on $W$.

*(a) The series $P_Q$ and $P'_Q$.* For a partition $\lambda$ let $b_\lambda(t) = \prod_{s\geq 1}\prod_{j=1}^{m_s(\lambda)}(1-t^j)$ [Mac, III, (2.12)], where $m_s(\lambda)$ is the number of parts equal to $s$. Note $\langle\lambda,(1)\rangle = l(\lambda)$, the number of parts. With $X^{|\lambda|} = \prod_i X_i^{|\lambda^i|}$, define

$$P_Q(X,u) = \sum_\lambda \frac{u^{\sum_{a\in Q_1}\langle\lambda^{t(a)},\lambda^{h(a)}\rangle}}{\prod_i u^{\langle\lambda^i,\lambda^i\rangle}\, b_{\lambda^i}(u^{-1})}\, X^{|\lambda|},$$

$$P'_Q(X,u) = \sum_\lambda \frac{u^{\sum_{a\in Q_1}\langle\lambda^{t(a)},\lambda^{h(a)}\rangle+\sum_i w_i\langle\lambda^i,(1)\rangle}}{\prod_i u^{\langle\lambda^i,\lambda^i\rangle}\, b_{\lambda^i}(u^{-1})}\, X^{|\lambda|}.$$

Since $\prod_i b_{\lambda^i}(u^{-1}) = (u^{-1})_{m(\lambda)}$ and $w_i\langle\lambda^i,(1)\rangle = \langle\lambda^i,(1^{w_i})\rangle$, the substitution $u = q^{-1}$, $X = z$ turns $P_Q$ and $P'_Q$ into the denominator and the numerator of the right side of (3.1).

*(b) Burnside's lemma and the Kac–Stanley formula.* For a finite $G_V$-set $R$, Burnside's lemma gives $|R/G_V| = |G_V|^{-1}\sum_{g\in G_V}|R^g|$. For $g \in G_V$, decompose each $V_i$ into the generalized eigenspaces $V_i^{(f)}$ of $g_i$, indexed by monic irreducible polynomials $f \neq t$ over $\mathbb{F}_u$. A linear map $\phi : V_i \to V_j$ with $g_j\phi = \phi g_i$ maps $V_i^{(f)}$ into $V_j^{(f)}$. Hence

$$R_Q(V)^g = \bigoplus_f R_Q(V^{(f)})^{g_f}, \qquad Z_{G_V}(g) = \prod_f Z(g_f),$$

so the Burnside sum factorizes over $f$. Consider the factor $f = t-1$, where the component of $g$ is unipotent. If $g_i - 1$ is nilpotent of Jordan type $\lambda^i$, then $R_Q(V)^g$ consists of the $\mathbb{F}_u[t]$-linear maps between the modules $V_i$, with $t$ acting by $g_i - 1$. Therefore

$$|R_Q(V)^g| = u^{\sum_a\langle\lambda^{t(a)},\lambda^{h(a)}\rangle}, \qquad |Z_{G_V}(g)| = \prod_i u^{\langle\lambda^i,\lambda^i\rangle} b_{\lambda^i}(u^{-1}),$$

and the unipotent elements of type $\lambda$ contribute $|R^g|/|Z_{G_V}(g)|$, the $\lambda$-term of $P_Q$. The same computation over $\mathbb{F}_{u^d}$ gives the local factor $P_Q(X^d,u^d)$ for $f$ of degree $d$, and one obtains the Kac–Stanley formula [Hua, Theorem 4.3]:

$$\sum_V |R_Q(V)/G_V|\, X^{\dim V} = \prod_{d\geq 1} P_Q(X^d,u^d)^{\varphi_d(u)},$$

where $\varphi_1(u) = u-1$ and, for $d \geq 2$, $\varphi_d(u)$ is the number of monic irreducible polynomials of degree $d$ over $\mathbb{F}_u$.

*(c) Why only eigenvalue 1 survives.* Since $G_V$ acts trivially on $W$, an element $g$ fixes a framing map $a_i \in \mathrm{Hom}(W_i,V_i)$ if and only if $g_ia_i = a_i$, i.e. $\mathrm{im}\, a_i \subseteq \ker(g_i-1)$. Thus

$$|\mathrm{Hom}(W_i,V_i)^g| = u^{\,w_i \dim\ker(g_i-1)},$$

which depends only on the component of $g$ at the eigenvalue 1. For $g_i$ unipotent of type $\lambda^i$, $\dim\ker(g_i-1) = l(\lambda^i) = \langle\lambda^i,(1)\rangle$. Adding the framing therefore changes only the local factor at $f = t-1$, from $P_Q$ to $P'_Q$:

$$\sum_V |R_Q(V,W)/G_V|\, X^{\dim V} = P'_Q(X,u)\, P_Q(X,u)^{u-2} \prod_{d\geq 2} P_Q(X^d,u^d)^{\varphi_d(u)}.$$

Dividing by the Kac–Stanley formula, all local factors at eigenvalues different from 1 cancel:

$$\frac{\sum_V |R_Q(V,W)/G_V|\, X^{\dim V}}{\sum_V |R_Q(V)/G_V|\, X^{\dim V}} = \frac{P'_Q(X,u)}{P_Q(X,u)}. \tag{3.2}$$

In [Nak04] this is derived from the Kac–Stanley formula for the quiver $Q^\#$ of Crawley-Boevey's trick, by taking the coefficients of $Y^0$ and $Y^1$, where $Y$ records the dimension at the extra vertex

$\infty$. There an element $(g,c)\in G_V\times\mathbb{F}_u^\times$ fixes the framing arrows $\infty\to i$ exactly when they land in the $c$-eigenspace of $g_i$. Since the diagonal scalars act trivially, $(g,c)$ and $(c^{-1}g,1)$ have the same fixed points, so the $u-1$ values of $c$ contribute equally; this factor $u-1$ cancels against $|GL_1(\mathbb{F}_u)|=u^{\langle(1),(1)\rangle}b_{(1)}(u^{-1})$. Normalizing $c=1$ gives the description above.

*(d) Krull–Schmidt and Crawley-Boevey–Van den Bergh.* A point $(B,a)\in R_Q(V,W)$ is a representation of $Q^\#$ with dimension 1 at $\infty$ [CB]. It decomposes uniquely as $(B_1,a_1)\oplus B_2$, where $(B_1,a_1)$ is indecomposable, hence absolutely indecomposable since the dimension at $\infty$ is 1, and $B_2$ is a representation of $Q$. Hence

$$\sum_V |R_Q(V,W)/G_V|\,X^{\dim V}=\sum_V |R_Q(V)/G_V|\,X^{\dim V}\times\sum_V |R_Q(V,W)^{\text{a.i.}}/G_V|\,X^{\dim V},$$

that is, framed = (absolutely indecomposable framed) × unframed. By [CBVdB, Proposition 2.2.1], in the form [Nak04, Theorem 1.1], for $W\neq 0$ and a generic parameter $\zeta\in\mathbb{F}_u^{Q_0}$ (for instance $\zeta=\eta$ when the characteristic exceeds $\sum_i v_i$),

$$|R_Q(V,W)^{\text{a.i.}}/G_V|=u^{-\delta(v,w)}\,\#\mathfrak{M}_\zeta(v,w)(\mathbb{F}_u).$$

Comparing with (3.2) gives $\sum_v u^{-\delta(v,w)}\#\mathfrak{M}_\zeta(v,w)(\mathbb{F}_u)\,X^v=P'_Q(X,u)/P_Q(X,u)$ [Nak04, Theorem 1.2], which is Hausel's formula. Finally $\#\mathfrak{M}_\zeta(v,w)(\mathbb{F}_u)=u^{2\delta(v,w)}P_{\mathfrak{M}_\eta(v,w)}(u^{-1})$, since the count is the compactly supported Poincaré polynomial of the complex quiver variety (see [Nak04, §2]), and the substitution $u=q^{-1}$ gives (3.1).

*(e) Loops.* If $Q$ has a loop at $i$, its fixed points $\mathrm{End}(V_i)^g$ form the centralizer algebra of $g_i$. For $g_i$ unipotent of type $\lambda^i$ it has $u^{\langle\lambda^i,\lambda^i\rangle}$ elements, which cancels the factor $u^{\langle\lambda^i,\lambda^i\rangle}$ in $|Z_{G_V}(g)|$. This is the cancellation of Lemma 3.4 below, seen from the Burnside side.

*Remark* 3.3 (Determinant stability). Let $X(v,w)$ be the quiver variety at determinant stability, i.e. the GIT quotient of $\mu^{-1}(0)$ by $G$ for the character det. For $\zeta=(\zeta_\mathbb{R},\zeta_\mathbb{C})\in\mathbb{R}^{Q_0}\oplus\mathbb{C}^{Q_0}$ let $\mathfrak{M}_\zeta(v,w)$ be the hyperKähler quotient of [Nak94, §2], with the convention $\mu=-\zeta$ used there. Then:

- by [Nak94, Theorem 3.2 and Proposition 3.5], $X(v,w)$ is homeomorphic to $\mathfrak{M}_{(\zeta_\mathbb{R},0)}(v,w)$ for a suitable nonzero real multiple $\zeta_\mathbb{R}$ of $(1,\dots,1)$. ([Nak94, Proposition 3.5] treats $\zeta_\mathbb{R}$ with positive entries, where stability is a condition on the maps $V\to W$; changing the sign of $\zeta_\mathbb{R}$ gives the dual condition, cyclicity with respect to the maps $W\to V$, which is the determinant stability used here.)
- by [Nak94, Theorem 3.1], $\mathfrak{M}_\eta(v,w)$ is homeomorphic to $\mathfrak{M}_{(0,-\eta)}(v,w)$, where $\eta=(1,\dots,1)$.

Both parameters lie off every wall $\mathbb{R}^3\otimes D_\theta$ with $0\neq\theta\in\mathbb{Z}_{\geq 0}^{Q_0}$, because $\sum_k\theta_k\neq 0$; so they are generic. By [Nak94, Corollary 4.2], $\mathfrak{M}_{(\zeta_\mathbb{R},0)}(v,w)$ and $\mathfrak{M}_{(0,-\eta)}(v,w)$ are diffeomorphic; in particular one is empty if and only if the other is. Hence $P_{X(v,w)}=P_{\mathfrak{M}_\eta(v,w)}$. This is the argument of [Hik, proof of Proposition 6.4].

The paper [Nak94] assumes that no edge joins a vertex with itself, whereas here $Q$ may have loops, and from §3.2 on it has a loop at every vertex. The proof of [Nak94, Corollary 4.2] does not use this assumption: it relies only on [Nak94, Theorem 4.1(1),(2)] and on rotating the complex structures $I,J,K$.

3.2. **Quivers with a loop at every vertex.** From now on $Q$ again has a loop at every vertex. Recall $Q'$ and $c_{ij}$ from (1.1), $C(v)$ from (1.2), and $l(m)$, $w(m)$, $(q)_m$, $D_m$ from Definition 1.2.

**Lemma 3.4** (Loop cancellation)**.** *For every $Q_0$-tuple of partitions $\lambda$, with $m=m(\lambda)$,*

$$E_Q(\lambda)=D_m,\qquad \sum_i\langle\lambda^i,(1^{w_i})\rangle=w(m),\qquad \delta(v,w)=C(v)+w\cdot v.$$

*Proof.* The chosen loops contribute $\sum_i \langle \lambda^i, \lambda^i \rangle$ to $\sum_{a \in Q_1} \langle \lambda^{t(a)}, \lambda^{h(a)} \rangle$, which cancels the vertex term; the remaining arrows give $D_m$ by (1.6). Next, $\langle \lambda^i, (1^{w_i}) \rangle = \sum_a \sum_{b=1}^{w_i} \min(\lambda^i_a, 1) = w_i\, l_i(m)$. The same cancellation applied to $\sum_a v_{t(a)} v_{h(a)} - \sum_i v_i^2$ gives $C(v)$. □

Using $\lambda \leftrightarrow m(\lambda)$, and with $h_m$ as in Definition 1.5, define the following series in $\mathbb{C}(q)[[z]]$:

$$\mathcal{A}^{\mathrm{fr}}(z) = \sum_{m \in \mathcal{M}} z^{\|m\|} \frac{q^{-D_m - w(m)}}{(q)_m}, \qquad \mathcal{A}^0(z) = \sum_{m \in \mathcal{M}} z^{\|m\|} \frac{q^{-D_m}}{(q)_m}, \tag{3.3}$$
$$\mathcal{H}(z) = \sum_{m \in \mathcal{M}} z^{\|m\|} q^{-D_m - w(m)} h_m(q).$$

Each $z^v$-coefficient is a finite sum over $\mathcal{M}(v)$, and $\mathcal{A}^0$ has constant term 1, so $\mathcal{A}^0$ is invertible. By Lemma 3.4 and Remark 3.3, Theorem 3.1 becomes

$$\sum_v q^{-C(v) - w \cdot v}\, P_{X(v,w)}(q)\, z^v = \frac{\mathcal{A}^{\mathrm{fr}}(z)}{\mathcal{A}^0(z)}. \tag{3.4}$$

### 3.3. **Computing the ratio by Proposition 6.2.**

**Corollary 3.5** ([Hik, Corollary 6.3])**.** *$\mathcal{A}^{\mathrm{fr}}(z) = \mathcal{A}^0(z) \cdot \mathcal{H}(z)$; equivalently $\mathcal{H} = \mathcal{A}^{\mathrm{fr}} / \mathcal{A}^0$.*

*Proof.* Recall the identity (6.1) of Proposition 2.1(b) and the series (3.3). Multiply (6.1) by $z^{\|m\|} q^{-D_m - w(m)}$. The left side becomes the $m$-th term of $\mathcal{A}^{\mathrm{fr}}$. For $k \le m$ with $m' = m - k$, (1.7) gives

$$-D_m - w(m) + w(k) + b(k, m') = -D_k - \big(D_{m'} + w(m')\big),$$

and $z^{\|m\|} = z^{\|k\|} z^{\|m'\|}$. So the $k$-th term on the right factors as

$$\left( z^{\|k\|} \frac{q^{-D_k}}{(q)_k} \right) \cdot \left( z^{\|m'\|} q^{-D_{m'} - w(m')} h_{m'}(q) \right),$$

a term of $\mathcal{A}^0$ times a term of $\mathcal{H}$. Since $(m, k) \mapsto (k, m - k)$ is a bijection from $\{(m, k) : k \le m\}$ onto $\mathcal{M} \times \mathcal{M}$, summing over all $m$ gives $\mathcal{A}^{\mathrm{fr}} = \mathcal{A}^0 \cdot \mathcal{H}$. □

Comparing the $z^v$-coefficients of (3.4) and Corollary 3.5, and multiplying by $q^{C(v) + w \cdot v}$, gives the formula of [Hik, (6.8)]:

**Corollary 3.6** ([Hik, Proposition 6.4])**.** *Let $X(v, w)$ be the quiver variety at determinant stability (Remark 3.3). Then*

$$P_{X(v,w)}(q) = \sum_{m \in \mathcal{M}(v)} q^{\,C(v) - D_m + w \cdot (v - l(m))}\, h_m(q).$$

### 3.4. **Examples.**

**Example 3.7** (Jordan quiver)**.** Let $Q$ be the Jordan quiver, so $Q'$ has no arrows and $c = 0$. Multipartitions are partitions $\lambda$, with $m_s$ parts of size $s$. The multipartition quiver $\Gamma_m$ has no arrows between slots, and every slot receives $w$ framing arrows. Hence $U_m = \prod_d (\mathbb{C}^w \setminus 0)$, $U_m / T_m = (\mathbb{P}^{w-1})^{l(\lambda)}$, $q_S = \prod_{d \in S} x_d^w$, $J_m = (x_d^w : d \in \Omega_m)$, and

$$h_m(q) = \prod_s \begin{bmatrix} m_s + w - 1 \\ m_s \end{bmatrix}_q,$$

the Poincaré polynomial of $\prod_s \mathrm{Sym}^{m_s}\mathbb{P}^{w-1}$. The Hesselink strata $S_{\lambda_S}$ (Corollary 2.3) are $\{a_d = 0\ (d \in S),\ a_d \neq 0\ (d \notin S)\}$, of codimension $w|S|$. Here $D_m = 0$, and every sum factors over part sizes. Euler's identity $\sum_{k\geq 0} t^k/(q)_k = \prod_{j\geq 0}(1-tq^j)^{-1}$ gives

$$\mathcal{A}^{\mathrm{fr}} = \prod_{s\geq 1}\prod_{j\geq 0}\frac{1}{1-z^s q^{j-w}}, \qquad \mathcal{A}^0 = \prod_{s\geq 1}\prod_{j\geq 0}\frac{1}{1-z^s q^j}, \qquad \mathcal{H} = \frac{\mathcal{A}^{\mathrm{fr}}}{\mathcal{A}^0} = \prod_{s\geq 1}\prod_{k=1}^{w}\frac{1}{1-z^s q^{-k}}.$$

Corollary 3.6 gives the Betti numbers of the Gieseker space $X(n,w)$:

$$P_{X(n,w)}(q) = \sum_{\lambda\vdash n} q^{\,w(n-l(\lambda))}\prod_s \begin{bmatrix} m_s + w - 1 \\ m_s \end{bmatrix}_q.$$

For $w = 1$ this is $\sum_{\lambda\vdash n} q^{\,n-l(\lambda)}$ for $\mathrm{Hilb}^n(\mathbb{C}^2)$; for $w = n = 2$ it is $(1+q+q^2)+q^2(1+q) = 1+q+2q^2+q^3$.

**Example 3.8** (The $A_2$ example continued)**.** In Example 1.13, $C(v) = 2$. For $m_b$: $D_{m_b} = 2$, $l(m_b) = (2,1)$, $w\cdot(v-l) = 0$, so the term is $q^0(1+2q)$. For $m_a$: $D_{m_a} = 1$, $l(m_a) = (1,1)$, $w\cdot(v-l) = 1$, so the term is $q^2\cdot 1$. Corollary 3.6 gives

$$P_{X((2,1),(1,0))}(q) = (1+2q)+q^2 = 1+2q+q^2,$$

for an 8-dimensional variety. Note that the unlooped $A_2$ variety with these data is empty; the formula concerns the quiver with loops.

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